CONCEPT NOTE

Title: Conversational AI and Intersectional Climate Risk Portfolio Explorer

Objective: Develop and implement a Climate Risk Portfolio Explorer, that integrates an AI conversational agent risk layer. The tool converts data on physical climate risks and social vulnerability into useful information presented in plain language for financial sector stakeholders, regulators, and portfolio managers.

Justification and relevance:

- Technical and scientific terminology: Data on climate risks (e.g., NGFS, Climate Analytics) are technically rigorous and are largely understandable only by specialists (scientists, climate experts, governments, etc.), making it extremely difficult for non-experts—such as civil society and the general public—to understand and reason about climate risks.
- Analytical and Synthesis Capabilities: Reports and dashboards on climate risks process large volumes of data, limiting the ability to conduct near-real-time analysis and synthesis to inform decision-making on climate risks.
- Beyond Physical Risk: Traditional financial climate risk tools measure physical hazards in isolation. The incorporation of ESG indicators guides analyses toward a better assessment of climate exposure, vulnerability, and risk.
- Institutional Alignment and SEI Innovation: The development of the Climate Explorer is fully aligned with SEI's mission to bridge the gaps between science, policy, and practice. By integrating generative AI and agent-based components, the tool aims to provide users with the ability to perform advanced queries, analyses, and searches with rapid and 24/7 assistance.

Methodology and architecture:

Stage 1: Climate Data Extraction

- Function: Automated download of climate data via API, calculated for Representative Concentration Pathway (RCP) scenarios 2.45 and 5.85.
- Variables or inputs: Extreme Heat Danger, Loss of Labor Productivity, Precipitation Adjustment, Extreme Rainfall over 5 Days, and Consecutive Dry Days.
- Advantage: Enables in-house access to climate data without relying on external sources or databases.

Stage 2: Creating and Structuring a GCP Cloud SQL Database

- Function: Creating a GCP cloud database using PostgreSQL, which organizes climate variables by RCP (2.45 and 5.85) based on coordinates (Lat/Lon). Additionally, the database integrates vector search and embedding fields to enable natural language and alphanumeric processing in AI.
- Advantage: Fast search of climate data for visualization on the web application's map, while enabling semantic search for the LLM via RAG.

Stage 3: Server and Virtual Machine in GCP Cloud (Generative AI and Agents)

- Function (Generative AI): Integration of the LLama3 (Open Source) Large Language Model (LLM) within the virtual machine, avoiding tokenization costs associated with external APIs and ensuring data sovereignty.
- Function (AI Agents): Orchestration of multi-agents using the AI LangGraph and Model Context Protocol (MCP).
    - ***Agent of Technical Analyst***: Extracts and synthesizes user query into a clean and structured LLM prompt following instructions and objectives.
    - ***Agent of Strategic Evaluator:*** Evaluate LLM response using technical instructions and objectives, reducing hallucinations and false information generated by LLM.
    - ***Agent of Create Reports****:* Create and prepare technical reports with climate data in PDF and Excel formats following instructions and templates.
    - ***Agent for Send Reports:*** Send Reports files with analysis (PDF and Excel) to user via online (e-mail or WhatsApp).
- Advantage: No rate limits, wait times, or token usage—users can make nearly unlimited queries at any time.

Stage 4: Handling user requests via RESTful API

- Function: Acts as an asynchronous communication bridge between the web application (Climate Explorer) and the virtual machine (generative AI and agents), allowing the user to send queries or instructions that are then forwarded or returned by the agents.
- Advantage: As a synchronous API (it first receives the user's request and then waits for the AI's response), it allows for handling multiple requests simultaneously without overloading the server and organizes responses in an interoperable format (JSON).

Stage 5: Domain, Security, and Web Deployment (Cloudflare + GitHub)

- Function: Domain Name System (DNS) and security certificate protocols using (Secure Sockets Layer – SSL and Transport Layer Security – TLS) via Cloudflare and GitHub actions for Continuous Integration (CI) and Continuous Deployment (CD).
- Advantage: Provides protection against Distributed Denial of Service (DDoS) attacks and secure routing/traffic over HTTPS.

Stage 6: UI/UX and Front-End Design

- Function: Main climate explorer phase that integrates User Interface (UI) and User Experience (UX) components, enabling interactive and dynamic user searches, queries, and filters related to climate risks.

Components (UI)

1. Modern and responsive design that adapts to various portable and PC devices.
2. Clean and attractive design, allowing for easy identification of panels, buttons, messages, etc.
3. Modular panel design, offering better layout and organization of information.

Components (UX)

1. Selection of base maps (Google Maps, ESRI Maps, OpenMaps, etc.).
2. Creating and drawing points, lines, or polygons on the map.
3. Importing and reading data or portfolio in CSV or TXT format.

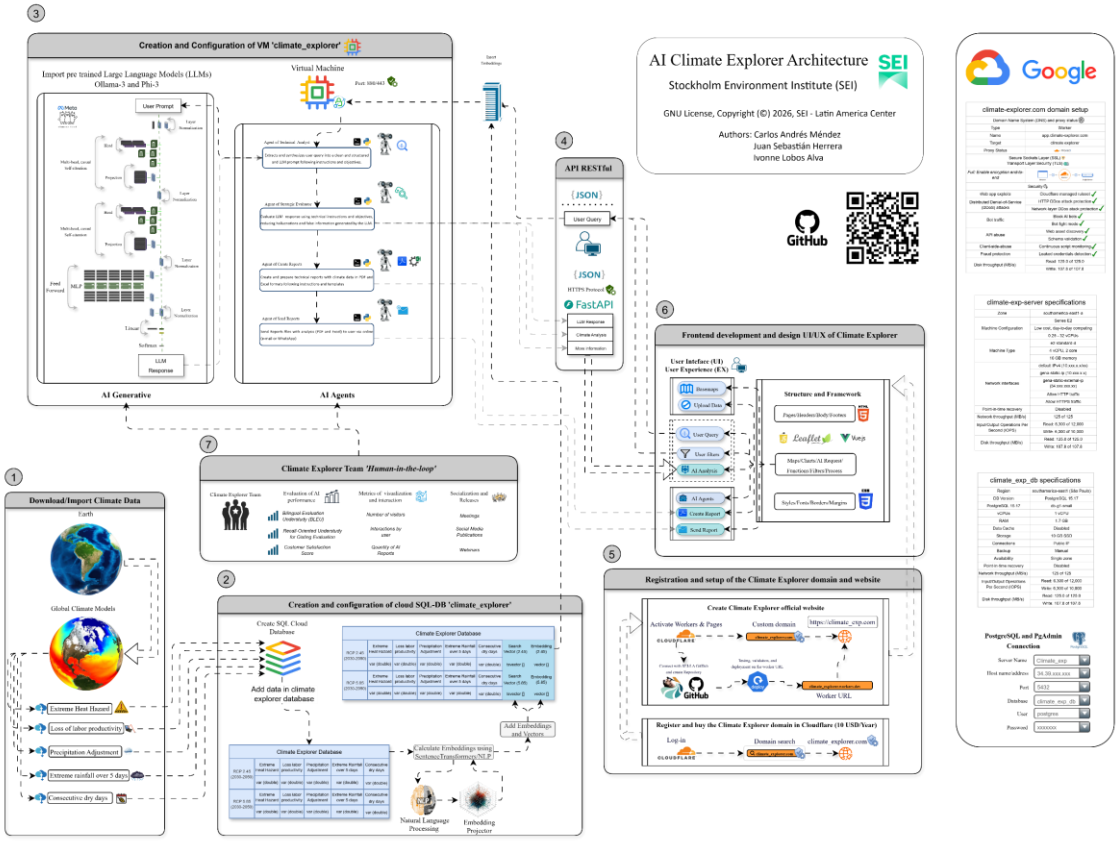
4. Selection of climatic variables for RCP scenarios 2.45 and 5.85.
5. Enabling a window for conversational chat with the AI
6. Download and send climate risk reports in PDF or Excel format.

- Advantage: Ultra-lightweight and fast design, eliminating the need to process large amounts of data directly within the web application.

Stage 7: Human-in-the-Loop Governance

- Function: Governance and ongoing review of the AI Climate Explorer by the SEI team, enabling them to evaluate the tool's performance, validate the AI performance, and receive feedback from users and climate risk specialists.
- Advantage: Proactively identify and resolve problems, errors, and inconsistencies in the tool's operation, thereby optimizing project time and resources.

AI Climate Explorer Tool v1.0 Architecture



End to End (E2E) Technological Stack

Backend | Database | Cloud Services | AI | Frontend

Estimated timeline:

SEI Team	Methodology and architecture of climate explorer		Time in months					
			2026					
Manager	Number	Phase or stage	July	August	September	October	November	December
Juan Sebastian Herrera	1	Climate Data Extraction	✔	✔				
Carlos Andres Méndez (30 hrs)	2	Creating and Structuring a GCP Cloud SQL Database			✔	✔		
Carlos Andres Méndez (30 hrs)	3	Server and Virtual Machine in GCP Cloud (Generative AI and Agents)				✔	✔	✔
Carlos Andres Méndez (30 hrs)	4	Handling user requests via RESTful API			✔	✔		
Carlos Andres Méndez (30 hrs)	5	Domain, Security, and Web Deployment (Cloudflare + GitHub)		✔	✔			
Juan Sebastián Herrera	6	UI/UX and Front-End Design	✔	✔				
Juan Sebastián Herrera	7	Human-in-the-Loop Governance					✔	✔